

VICTORIA BEACH, MB, Canada

WMO: 715520

Lat: 50.6944N

Lon: 96.5625W

Elev: 722

StdP: 14.32

Time Zone: -6.00 (NAC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.6	-18.3	-28.6	1.1	-22.4	-24.3	1.5	-18.2	31.9	9.5	29.2	8.1	7.0	320	0.730

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	82.5	70.8	79.8	69.4	77.3	68.2	73.4	79.5	71.4	77.5	69.7	75.6	10.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.4	119.1	77.0	69.3	110.6	75.3	67.4	103.6	73.7	37.5	79.7	35.7	77.2	34.2	75.7	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
31.0	27.3	24.3		-28.5	90.8	3.8	4.2	-31.2	93.8	-33.4	96.3	-35.5	98.6	-38.3	101.6
				-28.2	76.8	3.9	2.2	-31.0	78.4	-33.3	79.7	-35.5	80.9	-38.3	82.5

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	38.6	4.4	6.8	20.3	36.3	50.3	63.3	69.7	68.6	58.8	44.5	27.4	10.6	(6)
	DBStd	25.35	13.79	13.62	13.54	9.73	8.86	6.38	5.06	5.52	7.19	7.85	10.65	13.20	(7)
	HDD50	6189	1413	1211	921	416	106	2	0	0	12	208	677	1223	(8)
	HDD65	10021	1878	1631	1386	860	460	104	16	26	209	636	1127	1688	(9)
	CDD50	2027	0	0	0	6	115	402	611	578	276	38	1	0	(10)
	CDD65	383	0	0	0	0	4	54	163	139	23	0	0	0	(11)
	CDH74	1435	0	0	0	0	18	180	632	546	58	1	0	0	(12)
	CDH80	235	0	0	0	0	2	25	98	101	9	0	0	0	(13)
(14) Wind	WSAvg	11.7	11.1	10.4	10.7	10.6	11.1	11.1	11.2	12.0	13.2	14.0	13.4	11.4	(14)
(15) Precipitation	PrecAvg	21.5	0.9	0.6	0.9	1.1	2.5	3.6	3.1	2.9	2.3	1.7	1.1	0.9	(15)
	PrecMax	29.6	2.3	1.1	3.0	2.4	4.7	7.7	6.7	5.8	4.6	3.6	3.3	2.2	(16)
	PrecMin	14.1	0.3	0.0	0.1	0.4	0.7	1.5	0.7	0.6	0.7	0.4	0.2	0.2	(17)
	PrecStd	3.8	0.5	0.3	0.6	0.5	1.1	1.7	1.2	1.2	1.0	0.9	0.6	0.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.0	38.7	50.9	62.7	76.0	83.4	86.5	87.3	79.8	69.1	54.1	37.8	(19)
		MCWB	34.2	35.9	44.9	49.2	60.2	70.8	73.5	72.0	67.7	58.8	48.4	34.4	(20)
	2%	DB	32.5	35.1	44.1	57.2	70.3	78.6	82.5	82.5	74.9	62.9	47.4	34.0	(21)
		MCWB	31.0	33.3	39.0	46.1	57.7	67.0	71.7	70.3	65.5	55.5	43.3	32.2	(22)
	5%	DB	28.5	32.4	40.3	52.8	66.0	75.5	80.0	79.5	71.6	58.8	44.3	31.5	(23)
		MCWB	27.4	31.0	36.6	44.4	56.0	65.7	70.5	69.0	63.2	53.4	41.1	30.4	(24)
	10%	DB	24.3	27.3	37.0	49.0	62.4	72.8	77.5	76.8	68.6	55.4	40.5	28.5	(25)
		MCWB	23.4	26.0	34.5	42.2	54.2	64.0	69.3	67.9	61.9	50.8	38.3	27.6	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	34.7	36.5	45.6	50.3	63.8	73.7	76.8	75.5	70.3	60.0	48.9	35.0	(27)
		MCDB	35.7	38.3	50.3	59.1	70.5	79.9	82.3	82.0	76.2	66.2	53.4	37.4	(28)
	2%	WB	31.0	33.6	39.7	47.6	60.4	69.7	74.3	72.5	67.2	56.8	44.2	32.5	(29)
		MCDB	32.4	34.6	43.4	54.5	67.1	75.6	79.8	79.5	72.7	61.8	46.8	34.0	(30)
	5%	WB	27.2	30.9	36.9	45.2	57.7	67.3	72.4	70.6	65.0	53.8	41.2	30.3	(31)
		MCDB	28.3	32.4	39.9	52.4	63.9	73.2	77.9	77.4	70.0	58.0	43.9	31.5	(32)
	10%	WB	23.3	26.1	34.7	42.4	55.4	65.3	70.6	69.0	62.9	50.6	38.2	27.4	(33)
		MCDB	24.2	27.3	36.9	48.6	60.9	71.1	76.0	75.5	67.6	55.1	40.4	28.3	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	15.2	17.1	17.8	16.6	14.9	13.2	12.9	13.1	11.7	9.4	8.7	12.3	(35)
		MCDBR	15.1	13.5	16.2	20.6	19.8	17.3	15.8	17.5	16.1	15.2	11.9	12.7	(36)
		MCWBR	14.2	11.7	12.2	11.7	10.2	8.7	8.2	8.5	9.0	9.4	8.7	12.0	(37)
	5% WB	MCDBR	14.6	13.3	14.9	19.2	16.6	14.6	14.2	15.7	14.3	14.9	11.1	13.4	(38)
		MCWBR	13.7	11.6	11.5	11.6	9.6	8.9	8.9	9.0	9.1	9.6	8.3	12.4	(39)
(40) Clear-Sky Solar Irradiance	taub		0.265	0.288	0.295	0.348	0.361	0.381	0.390	0.374	0.344	0.300	0.272	0.248	(40)
	taud		2.223	2.174	2.323	2.361	2.404	2.371	2.373	2.425	2.494	2.574	2.436	2.289	(41)
	Ebn at Noon		242	266	289	284	284	278	273	272	270	266	244	233	(42)
	Edh at Noon		22	31	32	35	35	37	36	33	27	21	18	18	(43)
(44) All-Sky Solar Radiation	RadAvg		383	752	1160	1596	1730	1835	1911	1587	1123	650	348	278	(44)
	RadStd		38	54	86	108	147	144	89	112	102	83	49	28	(45)

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	-1.40	N/A	N/A	N/A	N/A
Regional (21 neighbors)	N/A	N/A	N/A	N/A	-1.06	-1.42	N/A	N/A	N/A	N/A

Nomenclature: See separate page